The total energy gained from the food = 200 food cal =200 kcal.

$= 200\times10^3 \times 4.186\ J = 8.37 \times10^5\ J$

If we assume that by using this energy the student can drawn 'n' pots of water from the well, the total energy spent by him = $8.37 \times 10^5\ J = nmgh$

$$n = \frac{8.37\times10^5 J}{6250 J} \approx 134.$$

This n is also equal to the number of trees that he can water.

Is it possible to draw 134 pots of water from the well just by having breakfast? No. Actually the human body does not convert entire food energy into work. It is only approximately 20% efficient. It implies that only 20% of 200 food calories is used to draw water from the well. So 20 % of the 134 is only 26 pots of water. It is quite meaningful. So he can water only 26 trees.

The remaining energy is used for blood circulation and other functions of the body. It is to be noted that some energy is always 'wasted'. Why is it that the body cannot have 100% efficiency? You will find the answer in section 8.9

## 8.6.2 First law of thermodynamics

The first law of thermodynamics is a statement of the law of conservation of energy. In Newtonian mechanics conservation of energy involves kinetic and potential energies of bulk objects. But the first law of thermodynamics includes heat also. This law states that '*Change in internal energy ($\Delta U$) of the system is equal to heat supplied to the system (Q) minus the work done by the system (W) on the surroundings*'. Mathematically it is written as

$$\Delta U = Q - W \qquad (8.13)$$

The internal energy of a thermodynamic system can be changed either by heating or by work as shown below.

| | |
|---|---|
| Heat flows into the system | Internal energy increases |
| Heat flows out of the system | Internal energy decreases |
| Work is done on the system | Internal energy increases |
| Work is done by the system | Internal energy decreases |

Based on the above table the sign convention is introduced to use first law of thermodynamics appropriately. It is shown in the following table and the Figure 8.20.

| | |
|---|---|
| System gains heat | Q is positive |
| System loses heat | Q is negative |
| Work done on the system | W is negative |
| Work done by the system | W is positive |

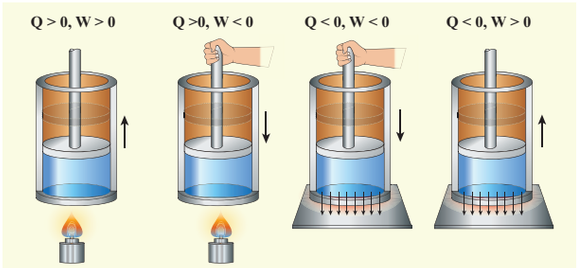



**Figure 8.20** The Sign convention for heat and work



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